

The UICC (USIM) standard is used when dealing with SIM card applications," adds Philippe Andre.

Consumer applications are mostly driven by a balanced alchemy between cost, performance and secured operating while the industrial or professional applications (EMV, healthcare, digital tachograph, game machine, identification) and, in particular, payments are driven by the need for an extremely safe and secured transaction systems. "The EMV systems are specific also by the fact that such a system can include up to five smart card interfaces and this has to be necessarily taken into account when defining the product," adds Philippe Andre.

Of course, semiconductor technology is not the only aspect of smart card technology that will change in the future. "On the packaging side too, significant progress has been made based on wafer thickness-reduction, die attach and wiring techniques and multi-chip packaging leading to reduced cost assembly processes. New R&D programs concentrate on heterogeneous material integration (sensors, polymer antennae) for improving the functionality and reliability of the components used in both contact and contactless smart card applications. In

With large consumer base and huge untapped potential, India offers immense opportunities to smart card players. The smart card industry has been continuously expanding, with states across the country identifying its true potential

cation to be a dominant application of smart card technology. "We anticipate that in the future we can expect M2M communication to be a dominant application of smart card technology," states Mohsin Syed.

Huge potential for manufacturers

Use of smart cards will be seen in almost all verticals in the years to come. Though India is still in the nascent stage of smart card adoption, it is witnessing tremendous growth. Considering the potential, the Smart Cards Manufacturers Association has been recently set up. This is a platform where members can inform, educate, advise and interact with private and government bodies about the technology. It works towards ensuring quality products and protects the manufacturers' interests.

"Currently, 50 per cent of the smart cards are being imported. But with the number of players setting up new

facturers do have the capacity to meet current as well as future demand. We must aim to work towards curbing or banning import of smart cards as it affects the growth of local manufacturers. There should be strict import duties levied so that Indian companies procure smart cards from local manufacturers and, hence, promote the smart card industry."

Industry experts therefore believe that smart card industry is a potential industry as it holds immense business opportunities. Further, the value and utility factor for a smart card solution is increasingly being noticed in the market with many scattered projects being initiated. "The growth of this segment will explode and make all paper work redundant," puts P.V. Bhas.

Smart cards are improving every year and have a steady future for at least nine to ten years until a new technology appears and replaces them. But for now, smart cards are the backbone of our digital life and personal identification.

"A typical day in the future will see smart cards being used by consumers to access and enjoy personalised educational and entertainment experiences, for civil and online identification purposes, to protect and monitor their health against unexpected threats, for access, transit, and payments, for faster and more convenient shopping experiences and much more. These cards will make life easier for us. Moreover, smart objects embedded in M2M applications will use heuristics to make our technological life simpler and better than ever before," signs off Mohsin Syed. ●

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the future, IP based solutions will enhance high-speed and high-bandwidth communications with greater security. Smart cards will act seamlessly and independently in an IP environment," informs Mohsin Syed.

Currently, the smart card industry has been addressing human-to-human communication and, in a limited way, human-to-machine communication. It has not yet addressed machine-to-machine (M2M) communication. But in future we can expect M2M communi-

facilities in India, we expect that more than 70-80 per cent of manufacturing will be done locally. The private sector is yet to contribute significantly to the growth of smart cards in India. However, the government has played a vital role in successfully building up the smart card industry in India. We expect quite a few more projects to be announced by the government, which will be mainly for the welfare of the citizens," says Anil Agarwal.

P.V. Bhas opines, "Indian manu-

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